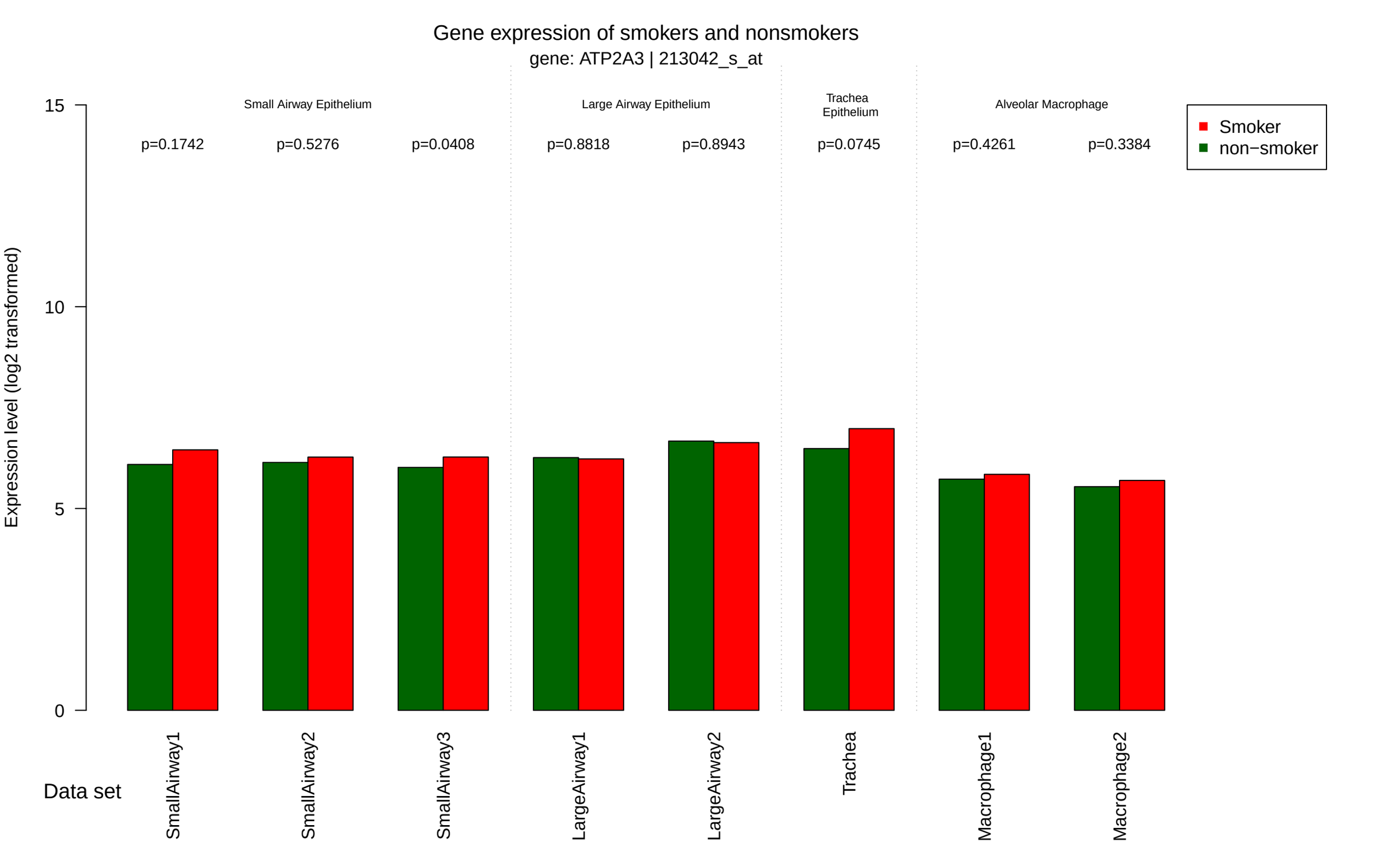
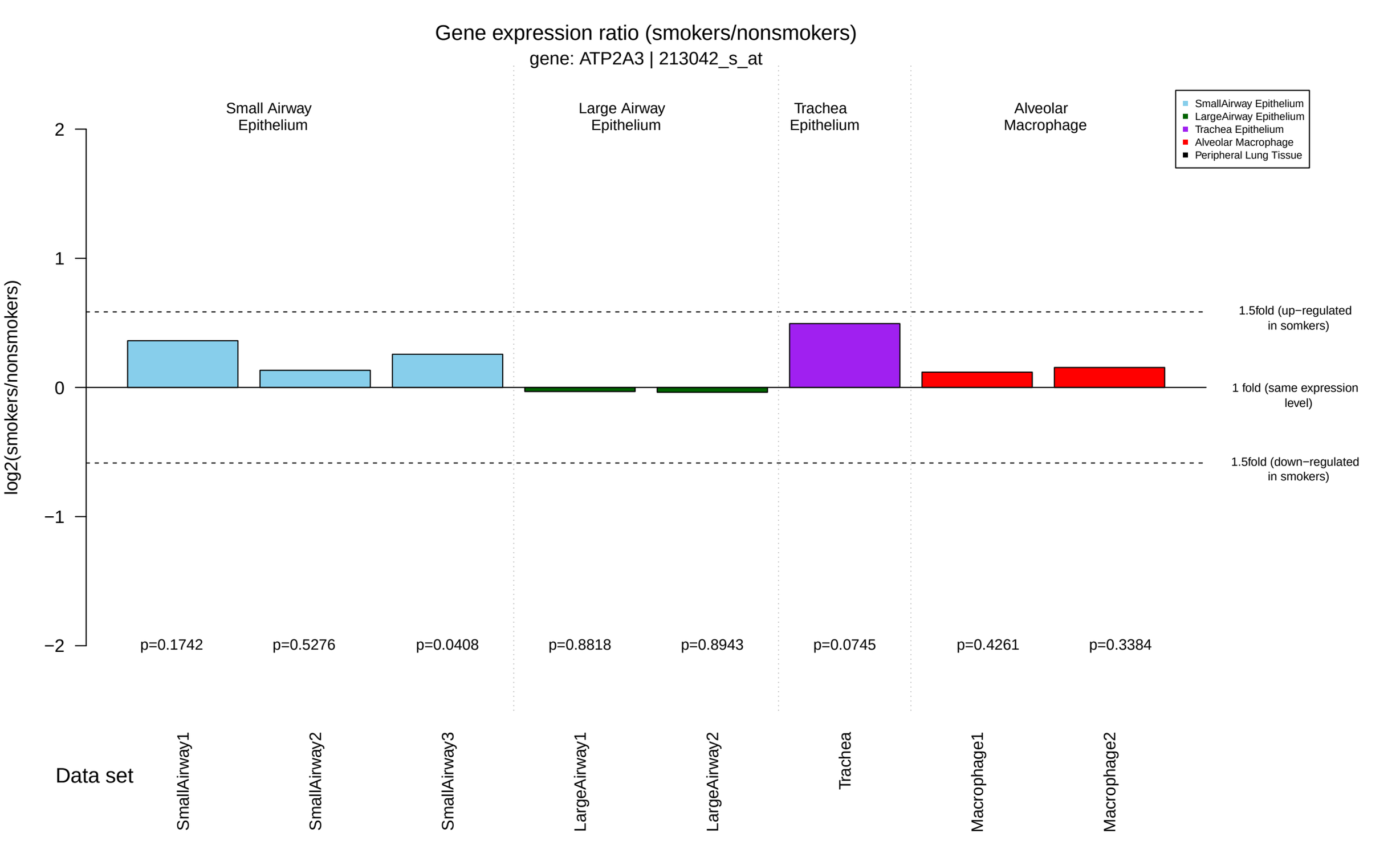
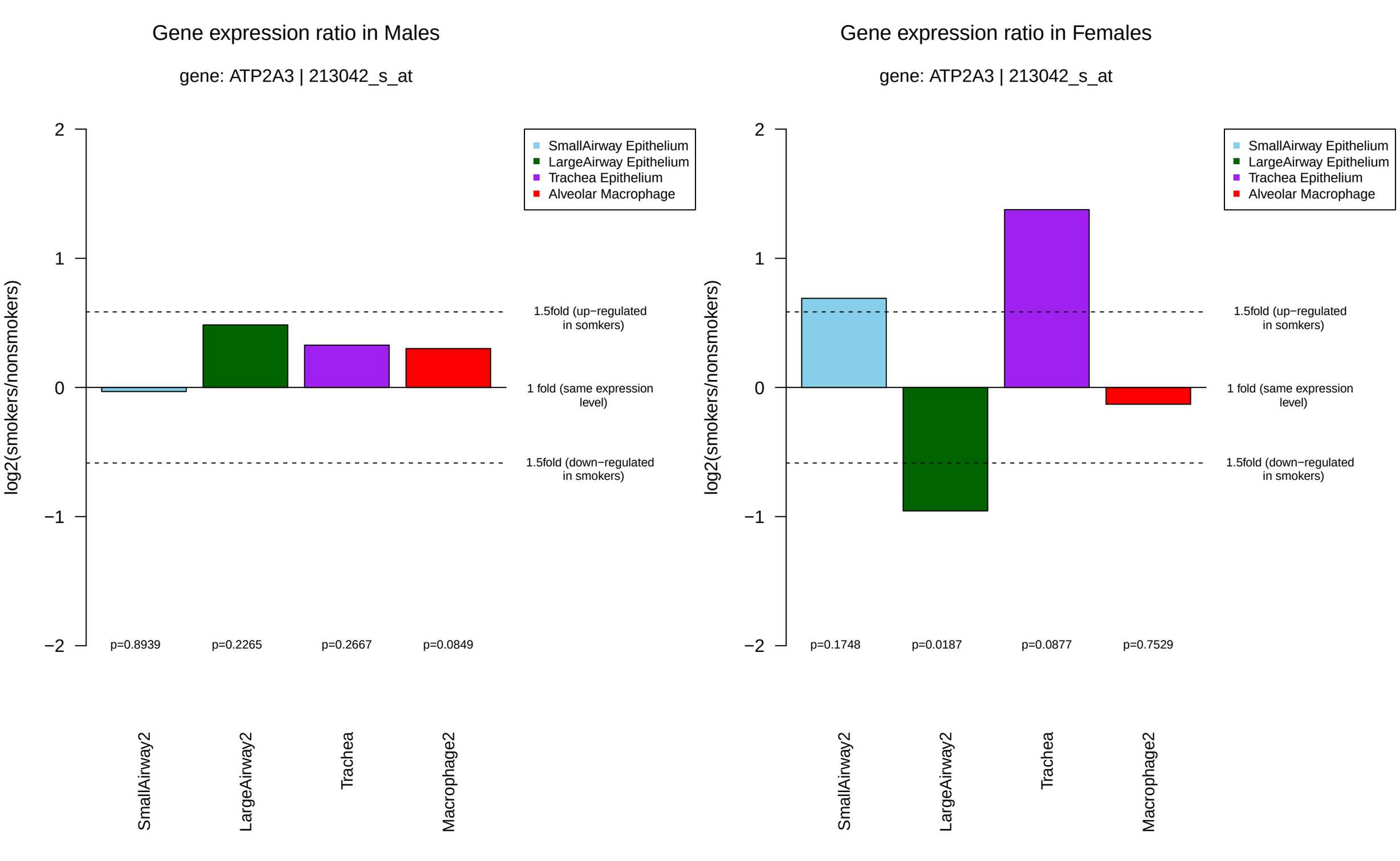
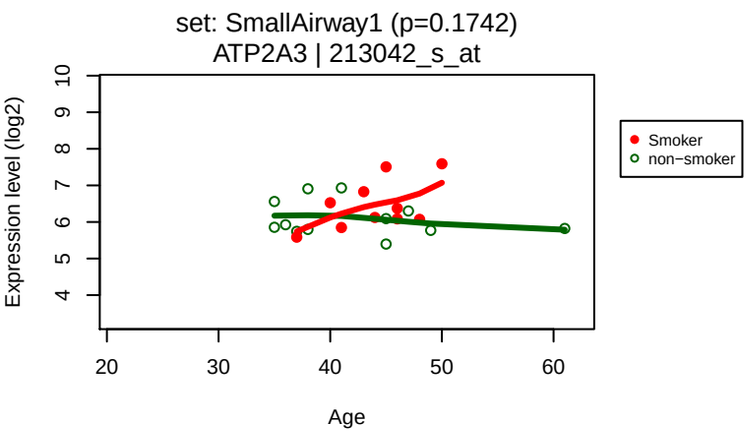
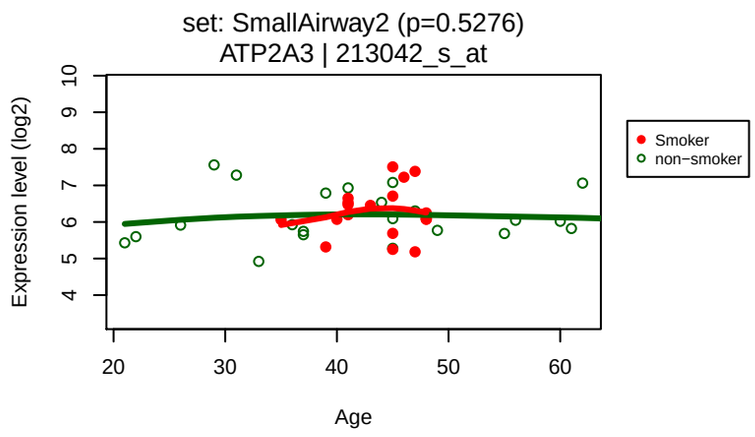
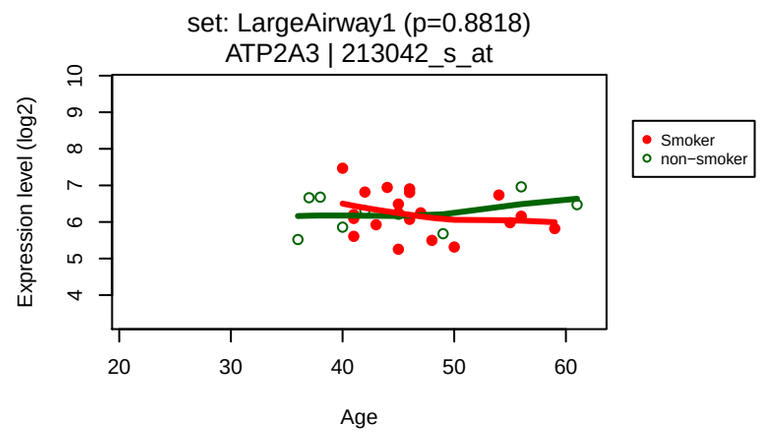

Gene expression of smokers and nonsmokers
gene: ATP2A3 | 213042_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1742
p=0.5276
p=0.0408
p=0.8818
p=0.8943
p=0.0745
p=0.4261
p=0.3384
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

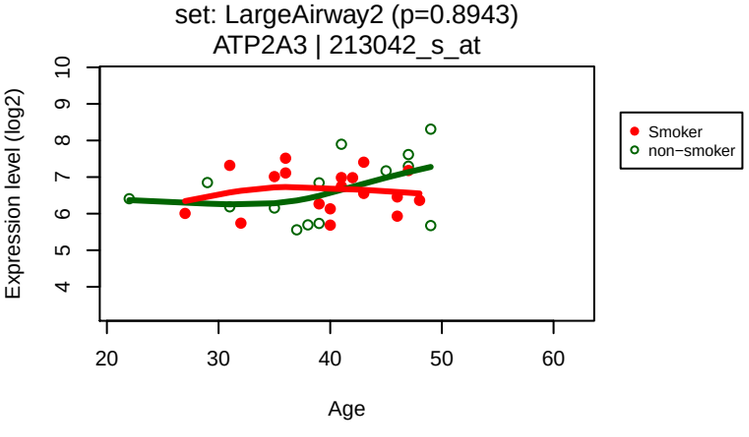

Gene expression ratio (smokers/nonsmokers)
gene: ATP2A3 | 213042_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1742
p=0.5276
p=0.0408
p=0.8818
p=0.8943
p=0.0745
p=0.4261
p=0.3384
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

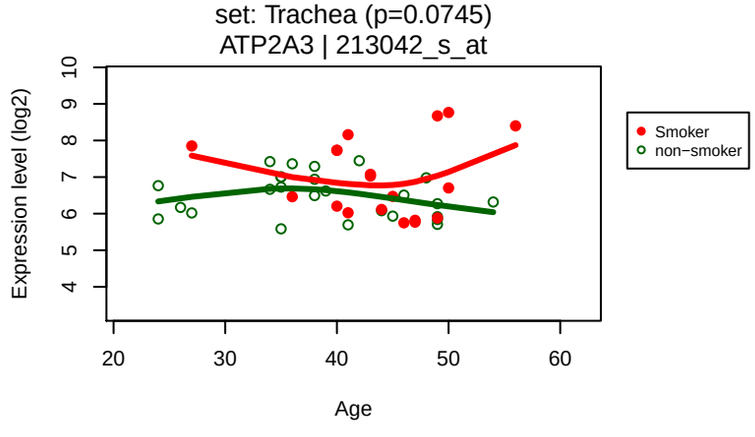

Gene expression ratio in Males
gene: ATP2A3 | 213042_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8939
p=0.2265
p=0.2667
p=0.0849
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP2A3 | 213042_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1748
p=0.0187
p=0.0877
p=0.7529
SmallAirway2
LargeAirway2
Trachea
Macrophage2

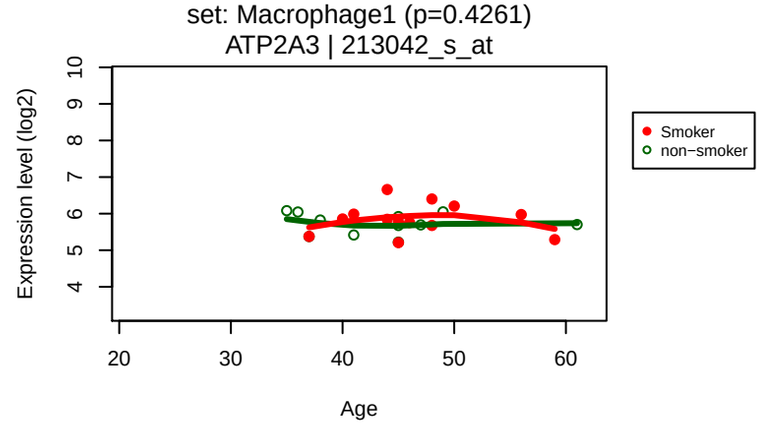

set: SmallAirway1 (p=0.1742)
ATP2A3 | 213042_s_at
Expression level (log2)
Age
Smoker
non−smoker

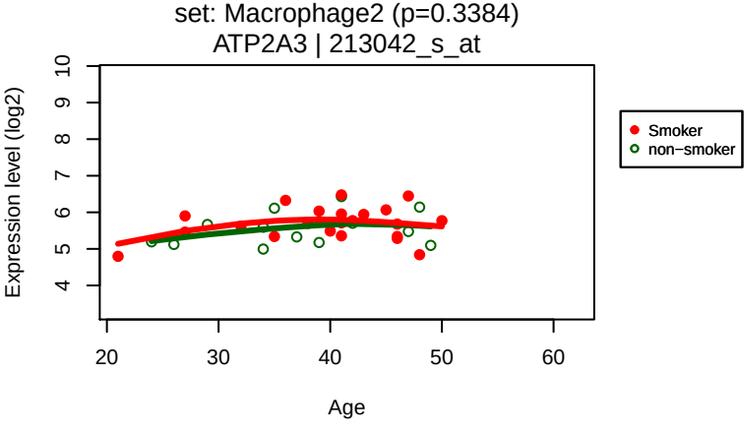

set: SmallAirway2 (p=0.5276)
ATP2A3 | 213042_s_at
Expression level (log2)
Age
Smoker
non−smoker

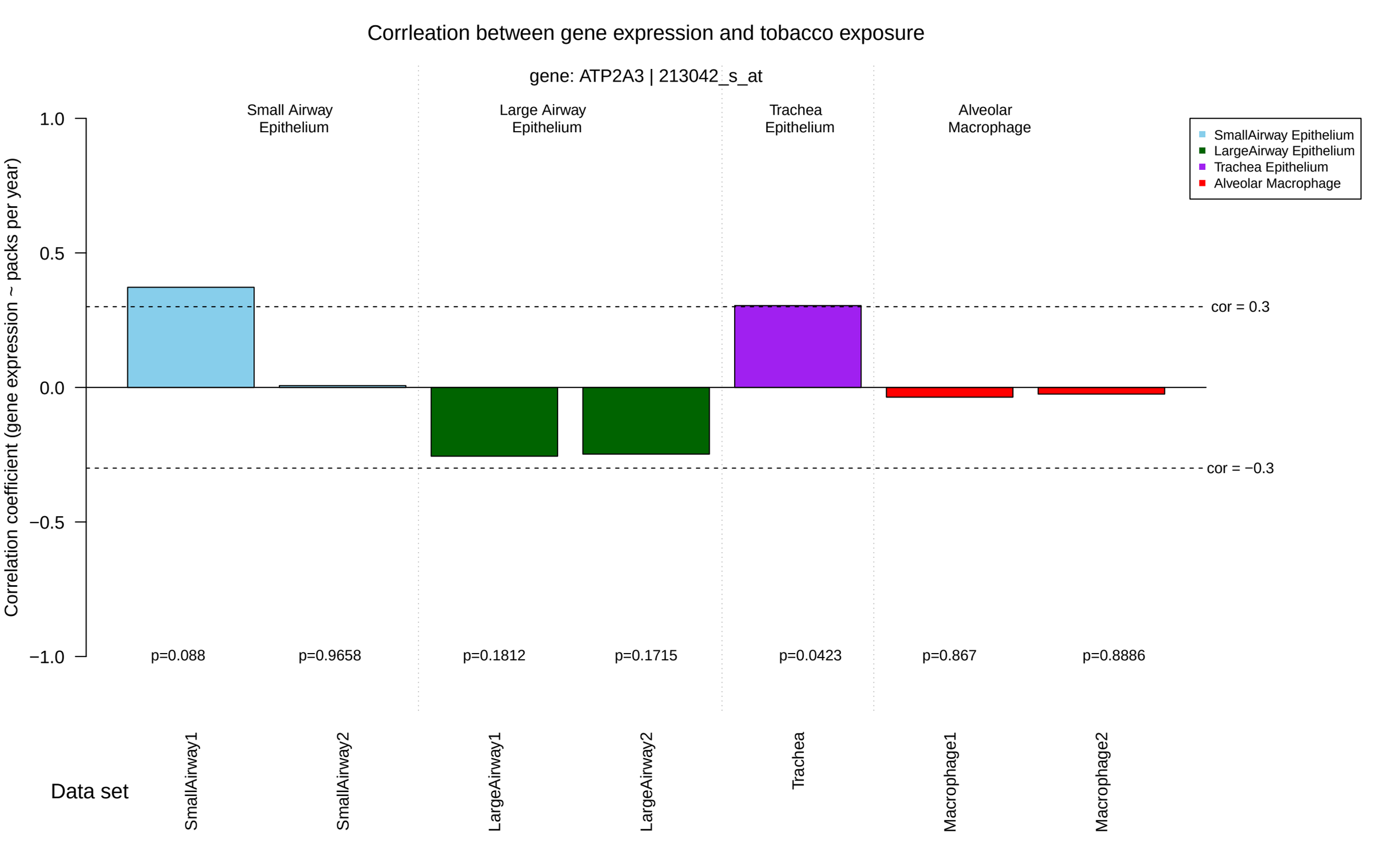

set: LargeAirway1 (p=0.8818)
ATP2A3 | 213042_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8943)
ATP2A3 | 213042_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0745)
ATP2A3 | 213042_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4261)
ATP2A3 | 213042_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3384)
ATP2A3 | 213042_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP2A3 | 213042_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.088
p=0.9658
p=0.1812
p=0.1715
p=0.0423
p=0.867
p=0.8886
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2